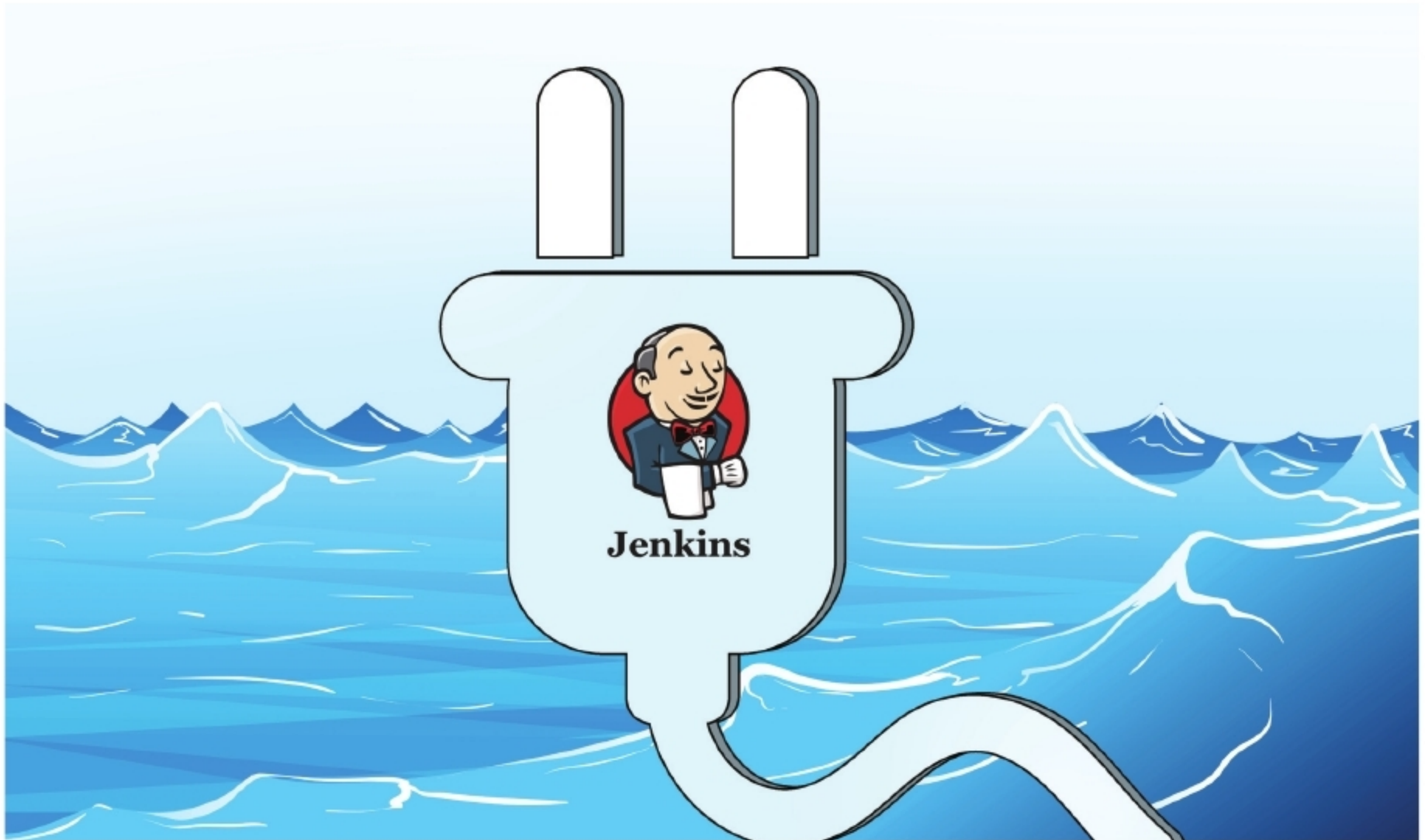


Pipeline as a Code: A Brief Look at Blue Ocean

Blue Ocean is a Jenkins plugin that enhances the user experience based on a personalised modern design. It has been built from the ground up to provide an interactive view of the Jenkins pipeline. It reduces the clutter and increases clarity for all users.



In Jenkins, pipelines used to be created using different plugins such as Build Pipeline or Build Flow, and so on.

However, after the Jenkins 2.0 release, pipelines (including the Blue Ocean plugin) have been integrated with the initial configuration.

The Build Pipeline plugin has been one of the most popular. It allows the creation of different styles of build jobs, and then connects all the jobs using upstream and downstream configurations. Once these configurations are completed, the Build Pipeline view can be created and the first job can be configured as a *Start* job in the Build Pipeline configuration. Once configured successfully, visualisation is available as per the relations configured in upstream and downstream jobs.

Pipeline as a Code

Pipeline as a Code was introduced using the scripted pipeline, which is an expression through which all activities

involved in application life cycle management can be automated using the script. A domain specific language is used to create the pipeline.

In Jenkins, a pipeline can be created using *Jenkinsfile*, which is a text file. If the name is different from the default one, then it has to be explicitly mentioned in the Pipeline configuration. This *Jenkinsfile* is committed in the source code repository such as Git or SVN.

Once the repository URL is provided, Jenkins searches for *Jenkinsfile* in each branch that is available in that repository. Once it finds this file in one or multiple branches, it starts execution.

You can filter branches based on wild cards to avoid multiple executions of pipeline code. A *Jenkinsfile* can be written using two types of syntax:

- Scripted pipelines
- Declarative pipelines